

CP-JKU SUBMISSION TO TASK 1 OF THE DCASE 2026 CHALLENGE: LLM PREDICTION FUSION AND PSEUDO-LABEL TRAINING FOR HETEROGENEOUS AUDIO CLASSIFICATION

Technical Report

Paul Primus¹, Gerhard Widmer^{1,2}

¹Institute of Computational Perception, Johannes Kepler University Linz, Austria

²LIT Artificial Intelligence Lab, Linz, Austria
first.last@jku.at

ABSTRACT

This report describes the CP-JKU submission to DCASE 2026 Task 1, Heterogeneous Audio Classification. The task requires assigning Freesound recordings to one of 23 second-level Broad Sound Taxonomy (BST) classes using audio and optional textual metadata. Our submitted systems use a diverse set of pretrained audio-only and audio-text backbones, including PaSST, BEATs, M2D, CP-CLAP, and LAION-CLAP. To exploit the metadata, GPT-5.4-mini is used in two complementary ways. First, it transforms the title, tags, and description into a concise caption-style summary that is encoded by the CLAP text encoder. Second, conditioned on the BST class definitions and representative examples, it predicts a probability distribution over plausible target classes from the unprocessed metadata and recording duration. This metadata-derived class prediction is mapped to a trainable metadata-prediction embedding and fused with the backbone representation. To exploit the larger but noisily annotated BSD35k-CS dataset, we use a three-stage procedure consisting of clean-data training on BSD10k, pseudo-labeled pretraining on BSD35k-CS, and final fine-tuning on BSD10k using the trusted ground-truth annotations. The submitted systems are ensembles of selected Stage-3 models and achieve up to 0.834 hierarchical F-score on our held-out BSD10k test split.

Index Terms— DCASE 2026, heterogeneous audio classification, Broad Sound Taxonomy, pseudo-labeling, metadata fusion, CLAP

1. INTRODUCTION AND TASK DESCRIPTION

DCASE 2026 Task 1 addresses heterogeneous audio classification with the two-level Broad Sound Taxonomy (BST) [1, 2]. Systems classify audio clips into one of 23 second-level sound categories. The development data consists of the carefully curated, cleanly annotated BSD10k dataset and the larger BSD35k-CS dataset with noisy annotations, both derived from Freesound audio clips [3, 2, 4]. The task also provides textual metadata, including title, tags, and description, which can be used by multimodal classification systems. The primary evaluation metric is macro-averaged hierarchical F-score [5], which gives partial credit to predictions that are correct at the top level of the taxonomy but incorrect at the second level. More details on the task setup, data, and evaluation protocol are provided on the official task website.¹

¹dcase.community/challenge2026/task-heterogeneous-audio-classification

Our submission focuses on two aspects of the task: exploiting textual metadata and using the noisy BSD35k-CS dataset.

- To exploit the metadata, we derive a class-probability prior using GPT-5.4-mini and fuse it with backbone embeddings using a learned class-embedding representation (Section 3).
- To deal with the noisy annotations in BSD35k-CS, we use pseudo-labels produced from Stage-1 predictions, followed by pretraining with those pseudo labels and fine-tuning on the clean BSD10k set (Section 4).

2. SYSTEM OVERVIEW

We train models based on five pretrained backbones. Three are audio-only backbones: PaSST [6], BEATs [7], and M2D [8]. Two are audio-text backbones: LAION-CLAP [9] with a tiny-HTSAT audio encoder [10] and a RoBERTa [11] text encoder, and CP-CLAP [12], a CLAP model based on PaSST and RoBERTa. For the audio-text models, the text input is a GPT-5.4-mini [13] summary generated from the available metadata fields. The prompt used to summarize the metadata is provided in our repository.²

3. METADATA PROCESSING AND LLM PREDICTION FUSION

The textual metadata associated with each Freesound recording consists of a title, a set of tags, and a free-form description. These fields provide a strong signal for BST classification, but they are heterogeneous in length and structure and may contain non-acoustic information such as licensing statements, attribution, links, HTML markup, or recording-device details. We therefore derive two complementary metadata representations: a continuous CLAP text embedding computed from an LLM-generated metadata summary and a metadata-prediction embedding derived from LLM-predicted BST class probabilities.

3.1. Metadata preprocessing and CLAP encoding

For each recording, GPT-5.4-mini [13] receives the title, tags, and description and produces a concise one- or two-sentence caption-style summary containing only information relevant to the expected

²github.com/OptimusPrimus/dcase2026-task1

sound content and BST category. The prompt emphasizes clear descriptions of audible events and removes promotional, licensing, attribution, and technical information. The summaries are generated offline and cached. The full prompt and resulting summaries are available in our repository.

Let s_i denote the resulting metadata summary for recording i . For the CLAP-based systems, the summary is encoded with the corresponding pretrained RoBERTa text encoder f_t^{CLAP} :

$$u_i = f_t^{\text{CLAP}}(s_i) \in \mathbb{R}^{D_t}, \quad (1)$$

where D_t is the text-embedding dimensionality. We use the encoder output before unit-norm normalization. For CP-CLAP and LAION-CLAP, this text representation is concatenated with the respective unnormalized audio representation to form the backbone representation h_i . For the audio-only PaSST, BEATs, and M2D systems, h_i consists only of the pooled audio representation.

3.2. Metadata-based class prediction with an LLM

In addition to the caption-style representation, we derive a representation that is explicitly aligned with the $C = 23$ second-level BST target classes. GPT-5.4-mini, with reasoning effort set to medium, is prompted to estimate a probability distribution over plausible BST classes using the unprocessed title, tags, description, and recording duration. The prompt contains the complete set of allowed class labels together with their definitions and representative examples. The response is required to be valid JSON containing class labels and probability values. The LLM may also assign probability to an auxiliary “other” class when none of the target definitions is appropriate. Because “other” is not a challenge target, it is removed and the remaining probabilities are renormalized to sum to one. Predictions are generated offline and cached.

Let

$$q_i \in \mathbb{R}^C \quad (2)$$

denote the resulting metadata-derived class prediction for recording i . Although q_i could be concatenated directly with the backbone representation, its dimensions have a fixed class-specific interpretation and its dimensionality is substantially lower than that of the backbone embeddings. We therefore map it to a continuous representation learned jointly with the classifier. Let

$$E \in \mathbb{R}^{C \times D_m} \quad (3)$$

be a trainable class-embedding matrix, where row E_c represents BST class c . The metadata-prediction embedding is the probability-weighted combination

$$m_i = q_i^\top E = \sum_{c=1}^C q_{i,c} E_c, \quad m_i \in \mathbb{R}^{D_m}. \quad (4)$$

This operation maps the class-aligned LLM output to a dense representation whose geometry is optimized for the final classification task.

3.3. Backbone-metadata fusion

The metadata-prediction embedding is concatenated with the backbone representation:

$$z_i = [h_i; m_i]. \quad (5)$$

For the audio-only systems, z_i combines the audio representation and the metadata-prediction embedding. For CP-CLAP and

LAION-CLAP, it combines the audio representation, the CLAP text embedding of the LLM-generated metadata summary, and the metadata-prediction embedding. The fused representation is processed by a two-layer fully connected network with GELU activation and dropout, followed by a linear classifier:

$$\ell_i = Wf(z_i) + b, \quad (6)$$

where $\ell_i \in \mathbb{R}^C$ contains the logits for the 23 target classes. A softmax is applied to obtain class probabilities.

4. PSEUDO-LABEL TRAINING WITH BSD35K-CS

BSD35k-CS contains substantially more recordings than BSD10k, but its crowdsourced labels are considerably noisier. To exploit the additional recordings without directly learning their annotation errors, we use soft pseudo-labels generated by an ensemble trained on clean BSD10k data. The resulting three-stage procedure consists of clean-data training, pseudo-labeled pretraining on BSD35k-CS, and final fine-tuning on BSD10k.

4.1. Stage 1: clean-data training

In the first stage, we train multiple multimodal classifiers on the BSD10k training split using the provided ground-truth class labels. The models use different pretrained audio backbones and therefore capture complementary acoustic information. For an input example i , let

$$\ell_{i,m} \in \mathbb{R}^C \quad (7)$$

denote the pre-softmax logits predicted by Stage-1 model m .

4.2. Stage 2: pseudo-labeled pretraining on BSD35k-CS

For each BSD35k-CS example, we average the logits of M selected Stage-1 models and apply a softmax transformation:

$$\tilde{y}_i = \text{softmax} \left(\frac{1}{M} \sum_{m=1}^M \ell_{i,m} \right). \quad (8)$$

The resulting vector $\tilde{y}_i \in [0, 1]^C$ is stored as a soft pseudo-label and used directly for training. In contrast to a hard one-hot target, the soft ensemble distribution retains information about the relative plausibility of multiple classes [14].

Each BSD35k-CS example has two potential supervision signals: its original crowdsourced one-hot label and the soft pseudo-label predicted by the Stage-1 ensemble. We compute a cross-entropy loss against each target and combine them as

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{\text{supervised}} + \lambda\mathcal{L}_{\text{pseudo}}, \quad (9)$$

where $\lambda \in [0, 1]$ controls the influence of the teacher ensemble. For the challenge systems, λ is sampled from $[0.75, 1.0]$ and tuned separately for each backbone. Higher values generally perform better, indicating that the Stage-1 ensemble predictions provide more reliable supervision than the original crowdsourced annotations for many recordings.

Table 1: Hyperparameter search ranges. Learning rates are sampled log-uniformly; pseudo-label weights λ are sampled uniformly.

Backbone	Stage	LR range	Epochs / λ
PaSST	1	$5 \cdot 10^{-6}$ – $1 \cdot 10^{-5}$	30 / –
	2	$3 \cdot 10^{-6}$ – $6 \cdot 10^{-6}$	15 / .75–1.0
	3	$0.9 \cdot 10^{-6}$ – $2 \cdot 10^{-6}$	15 / –
BEATs	1	$3 \cdot 10^{-6}$ – $6 \cdot 10^{-6}$	15 / –
	2	$3 \cdot 10^{-6}$ – $6 \cdot 10^{-6}$	15 / .75–1.0
	3	$1 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	15 / –
M2D	1	$1 \cdot 10^{-6}$ – $4 \cdot 10^{-6}$	30 / –
	2	$1 \cdot 10^{-6}$ – $3 \cdot 10^{-6}$	30 / .75–1.0
	3	$1 \cdot 10^{-6}$ – $3 \cdot 10^{-6}$	15 / –
CP-CLAP	1	$1 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	20 / –
	2	$2 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	15 / .75–1.0
	3	$2 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	15 / –
LAION-CLAP	1	$1 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	20 / –
	2	$2 \cdot 10^{-6}$ – $5 \cdot 10^{-6}$	15 / .75–1.0
	3	$1 \cdot 10^{-6}$ – $4 \cdot 10^{-6}$	15 / –

4.3. Stage 3: clean-data fine-tuning

In the final stage, the Stage-2 models are fine-tuned on BSD10k using only the original ground-truth annotations and the standard supervised cross-entropy objective. This step readjusts the decision boundaries toward the trusted BSD10k label distribution after pre-training on the larger pseudo-labeled dataset. It is intended to preserve useful representations learned from BSD35k-CS while correcting biases caused by residual pseudo-label errors or differences between the BSD10k and BSD35k-CS data distributions.

5. TRAINING AND MODEL SELECTION

We split BSD10k into training, validation, and test subsets, corresponding to one fold of the task baseline setup. The validation set is used for checkpoint selection, hyperparameter selection, and ensemble construction. The results reported in Section 7 are computed on the held-out BSD10k test split. BSD35k-CS is used only for training in Stage 2 and is not used for validation, model selection, or testing.

Hyperparameters are tuned separately for each backbone and training stage. The learning rate is tuned in all stages, while the pseudo-label loss weight λ is used only in Stage 2. The ranges are shown in Table 1. For every backbone-stage combination, we run 20 randomly sampled configurations and select models based on validation-set hierarchical F-score. The batch size is fixed to 18 for all experiments. The learning rate schedule uses linear warmup followed by linear decay to 10^{-7} until the final epoch. We use AdamW [15] with PyTorch’s [16] default momentum parameters and fix the weight decay to 0.01. We further use early stopping based on validation hierarchical F-score. Learning rates are sampled log-uniformly, and the Stage-2 pseudo-label weights are sampled uniformly. We apply class-frequency loss weighting to counteract class imbalance in BSD10k.

6. SUBMITTED SYSTEMS

We submit four final systems. Each system is an ensemble of selected Stage-3 models. In all systems two CP-CLAP models are used, one CP-CLAP model uses LLM prediction fusion (Section 3) and one does not. The no-fusion CP-CLAP model still uses the RoBERTa text branch with the GPT-generated metadata summary (Section 2), but it does not use the additional LLM prediction embedding.

- **System 1** combines BEATs, two CP-CLAP models, and one LAION-CLAP model. It uses BEATs, CP-CLAP PaSST, and LAION-CLAP tiny-HTSAT audio representations, CP-CLAP and LAION-CLAP RoBERTa text representations, and LLM prediction embeddings where enabled.
- **System 2** combines two CP-CLAP models and one M2D model. It uses CP-CLAP PaSST and M2D audio representations, CP-CLAP RoBERTa text representations, and LLM prediction embeddings where enabled. The ensemble has 383M parameters.
- **System 3** uses the same model types and representation types as System 2, but with a different set of trained model instances.
- **System 4** combines two CP-CLAP models, one LAION-CLAP model, and one M2D model. It uses CP-CLAP PaSST, LAION-CLAP tiny-HTSAT, and M2D audio representations, CP-CLAP and LAION-CLAP RoBERTa text representations, and LLM prediction embeddings where enabled.

7. RESULTS

Table 2 reports the performance of the four submitted systems on our custom BSD10k development split, which imitates the official baselines split. System 3 obtains the highest hierarchical F-score on this split.

Table 2: Performance on our BSD10k test split.

System	hP	hR	hF
System 1	0.830	0.831	0.830
System 2	0.831	0.837	0.832
System 3	0.833	0.838	0.834
System 4	0.828	0.832	0.829

8. CONCLUSION

We presented a submission to DCASE 2026 Task 1 based on pre-trained audio and audio-text backbones, LLM-based metadata processing, and pseudo-label training for noisily annotated data. The metadata component derives a metadata-based class prediction from the textual metadata, maps it to a trainable metadata-prediction embedding, and fuses it with the backbone representation. The training procedure uses ensemble pseudo-labels to exploit BSD35k-CS while reducing the influence of noisy crowd-sourced annotations. Final predictions are obtained by ensembling selected Stage-3 models.

9. ACKNOWLEDGMENT

The LIT AI Lab is supported by the Federal State of Upper Austria. GPT-5.5 was used to assist in drafting and editing this technical report.

10. REFERENCES

- [1] DCASE Community, “DCASE 2026 Challenge Task 1: Heterogeneous Audio Classification,” <https://dcase.community/challenge2026/task-heterogeneous-audio-classification>, 2026, accessed: 2026-06-16.
- [2] P. Anastasopoulou, J. Torrey, X. Serra, and F. Font, “Heterogeneous Sound Classification with the Broad Sound Taxonomy and Dataset,” in *Proceedings of the Detection and Classification of Acoustic Scenes and Events Workshop*, Tokyo, Japan, Oct. 2024.
- [3] F. Font, G. Roma, and X. Serra, “Freesound Technical Demo,” in *Proceedings of the 21st ACM International Conference on Multimedia*. Association for Computing Machinery, 2013, pp. 411–412.
- [4] P. Anastasopoulou and F. Font Corbera, “BSD35k-CS (Broad Sound Dataset 35k - Crowd Sourced),” Mar. 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.19187100>
- [5] P. Anastasopoulou, F. A. Dal Rí, X. Serra, and F. Font, “Hierarchical and Multimodal Learning for Heterogeneous Sound Classification,” in *Proceedings of the Detection and Classification of Acoustic Scenes and Events Workshop*, Barcelona, Spain, Oct. 2025.
- [6] K. Koutini, J. Schlüter, H. Eghbal-zadeh, and G. Widmer, “Efficient Training of Audio Transformers with Patchout,” in *Proceedings of Interspeech*, 2022, pp. 2753–2757.
- [7] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, “BEATs: Audio pre-training with acoustic tokenizers,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 5178–5193.
- [8] D. Niizumi, D. Takeuchi, Y. Ohishi, N. Harada, and K. Kashino, “Masked Modeling Duo: Learning representations by encouraging both networks to model the input,” *arXiv preprint arXiv:2210.14648*, 2022.
- [9] Y. Wu, K. Chen, T. Zhang, Y. Hui, M. Nezhurina, T. Berg-Kirkpatrick, and S. Dubnov, “Large-Scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2023.
- [10] K. Chen, X. Du, B. Zhu, Z. Ma, T. Berg-Kirkpatrick, and S. Dubnov, “HTS-AT: A hierarchical token-semantic audio transformer for sound classification and detection,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022.
- [11] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [12] P. Primus, F. Schmid, and G. Widmer, “Estimated Audio-Caption Correspondences Improve Language-Based Audio Retrieval,” in *Proceedings of the Detection and Classification of Acoustic Scenes and Events Workshop*, Tokyo, Japan, Oct. 2024, pp. 121–125.
- [13] OpenAI, “GPT-5.4 mini Model Documentation,” <https://developers.openai.com/api/docs/models/gpt-5.4-mini>, 2026, accessed: 2026-06-16.
- [14] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015.
- [15] I. Loshchilov and F. Hutter, “Decoupled Weight Decay Regularization,” in *International Conference on Learning Representations*, 2019. [Online]. Available: <https://openreview.net/forum?id=Bkg6RiCqY7>
- [16] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.